ActInf Textbook Group ~ Cohort 6 ~ Session 16 (Chapter 7, Part 1) 8/5/2024

TextbookGroup/ParrPezzuloFriston2022/Cohort_6 | Cohort 6 ~ Session 16 (Chapter 7, Part 1) 8/5/2024 | 1:00:20

all right welcome everyone
from Co cohort 6 and 2/3 um we're in
chapter 7
discussion and then also in cohort 7
were in Chapter 2 so we'll maybe explore
a little bit of both of them and also
I'll just note that the octopus math
sessions are in Full Effect they're
Tuesdays one hour earlier than this um
the zoom link is
here and they're they're doing really
fun the recordings are available and
they're doing
fun math education in octopus is super
knowledgeable so if anyone has questions
on from from very basic background math
to more speculative maths I recommend
they look at the recordings or go to
those
sessions um all right
so
on seven or or to if if uh it's relevant
what's something that anyone would like
to begin
with so for the discretization of time I
feel like there were a few things that I
wanted to be a little bit more defined
could you say a little bit about how you
actually take this thing that the
discretization of time or States seems
like it could be broken up into smaller
and smaller pieces which kind of leads
me to believe that there's some bundling
that's going on behind the scenes and it
makes me feel like I want this

discretization as well as the definition
of time to be brought into in a little
bit more depth could you say a little
bit about your thoughts about
that yes great
question so there's a few different
settings where you might be applying a
discrete time model first let's let's
just go to figure
4.3 kind of the Rosetta Stone with the
disree and the continuous time so this
image highlights that with the same
structural relationship amongst
different variables we can have a
discreet time or a continuous time
active inference setting so that's kind
of handled chapter seven and eight
that's what a big part of the second
part of the book is so H how is time
discretized is this a claim about real
time is this like an ontological Claim
about time or how does it come into play
instrumentally so sometimes you may be
interested in making an example where
there actually is a discrete time
counter like it's a metronome or a clock
or you're using minutes or hours so
again whether or not you took a
ontological position that time
underlying the minutes was continuous if
you were measuring by the minute you
whether you're doing trading or or some
other kind of measurement you could
still make a discrete time just by
having um some kind of
click um in the mouse te's example it's
even less problematic I think
because there's no clock time wall time

it's just action time so it's like
action State step one and then an action
is taken so that's kind of like Chess
Time where it's disree and there's not
even a reference to a wall clock time so
that's kind of the simplest setting is
where there's no clock it's just like a
chess game then the sort of intermediate
setting is real time is happening but
your modeling discreet units of time
like daily or hourly and then the sort
of most open and challenging of the
settings relatively is where you have a
degree of Freedom around how you
discretize the time like what should the
time discretization be and so there that
is basically getting into a parameter
sweep over
possible Delta
T's and then that gets into okay so now
we have multiple possible models some
are taking at a finer scale some are
taking at a slower scale and then do we
get like more accuracy or
interpretability or are the benefits in
accuracy and interpretability worth the
increased cost for a faster ticking
model
so do you mind if I just try and
reiterate this again so the question was
what uh how do we make the choice about
the uh the sort of time that we're going
to be modeling modeling in the discrete
sense so we're gonna have either some
absolute clock time or some purely
ordinal sense of events
um you know following one another and
the question is like how do we choose

between
those because I think if that is the
question uh it would be very use case
dependent so you know in in something
where there's a lot of interacting
components you're going to have to be
aware of conflicts between events that
can happen but if if you're just
modeling a mouse and a maze maybe you
can just have a global clock time and
not have to worry about the um the
clashes of events that's that's the only
reason I I bring up
that yeah good good and yeah here
there's no clock at all it's just it's
time point one and then an action is
taken and then it's time point
two right sorry that's what I meant you
don't you don't need a clock because you
don't have to worry about
clashes so I suppose I suppose it would
be use case dependent um the choice
let's
say like there's nothing a priori in
active inference that would perhaps
necessitate one choice or the
other I I at least I don't see
how um in the chapter 6 recipe I mean
this is a yeah go ahead well I just me
uh let's say you're you're trying to
just decide which of those
two you know kinds of of times you want
to have without the specific use case
that is to say this specific example
like there there's there I'm trying to
think like there's nothing that would
mitigate in favor of one or the other
like it's very use case dependent I I

think I could be wrong though yeah I
mean in the particulars every single
aspect of the generative model is system
or use case specific there's there's
nothing that you could reduce your
uncertainty about further without
reference to a particular
system yeah
yeah like even in the in the chapter 6
recipe that's one of the questions is
you know what form should the model have
should we use discrete time discrete
State space continuous time continu spad
base or or hybrids of
them so at the at the system independent
scale we're just learning the motifs and
the possible moves and then they're all
then then they're selected from and
composed in making generative model for
a specific situation so like in chapter
7 it starts off
familiarizing with the discrete time
partially observable downstairs sense
making part of hidden State temperature
in the room observation thermometer
readings or in the in the chapter 7 case
it's the true note that was meant to be
played and then this is the observed
note that's listened
to D is your beginning prior on hidden
state it just kicks off this Markoff
chain and then after that it the the
prior for the following hidden state is
just the preceding hidden State a maps
from hidden states to
observations so it can be run forward in
the so-called generative direction again
it's a little confusing with the

namespace because the whole thing is a
generative model but it's generating
data synthetic data like generating
observations conditioned upon a
temperature of the room or in the
recognition Direction taking in
observations from the thermometer and
then making inference about what the
likely hidden state of the room is
and then B is a transition operator so s
is hidden State at a given time B is the
discrete time State update function then
you get hidden State at the next time
point so this is kind of the the
downward facing e that ends up being the
downstairs of the
PDP so this highlights just the part
partially observable
component and then action is brought in
later and the example in chapter 7 is
listening to
music so we can look more at that
example or we can go any anywhere or any
question that someone
prefers I I have a question I suppose
maybe following on from that PDP with
with action so I think that's figure 7.3
if I'm not mistaken but um if anyone
else has something they want to jump in
with please don't let me stop you so
yeah so 7.3 here um there's a bit of a
discussion about this you know we've
introduced action now uh well we've
introduced policies now you know very
nice and this this is um we we need the
EF to score these policies the question
that that I have is about and this is
not really I don't really see here in

this chapter where it's kind of finessed but on the next page I don't know if that's relatively easy to do equation 7.4 they actually give expect free energy um and of course the you know within the various terms you've got Pi your policy um just kind of um you know interpreting this Pi Pi is a sequence of actions so there is in my mind let's say you know I'm trying to think how do I model some particular um event or you know system I don't it's going to be kind of bad to have to try and form uh approximate posterior Bel over sequences of actions that's going to get quite massive very quickly you know uh you're going to grow you're going to have to form exponentially many approximate posterior beliefs for you know one more uh action that you add to your policy so there there arises pretty quickly this necessity to deal with that problem um and I'm just wondering if in in the text uh or in general if there is a relatively uh let's say kind of first pass you know way we can deal with this problem um the one that comes to mind immediately is to just assume that the um the approximate posterior over policies is you can just do a mean field thing so instead of having to do you know beliefs over things that are 16 times into the future you just say okay I'm going to assume that I can multiply the probability of action one versus action two versus action three and form posterior beliefs over that

over you know individual actions and
sort of just you know take the product
um but nevertheless this this is
something that is a little bit more of a
general uh query that I've had for a
long time about the EF you know you have
to to kind of you're sort of forced to
deal with this problem of how to uh
represent beliefs over policies because
those policies can get so enormous very
quickly so I'm sorry that's maybe not
too uh well formed but alas that is my
question if anyone wants to jump
in yeah does anyone want to give any
thoughts or or is is it related to
anything someone's thought or wondered
and then let's explore
it um yeah so I I've actually seen
something somewhat similar to this
question recently
um where you know like the naive way is
to do Monte Carlos sampling right and
you just you looking over like you said
like every possibility in this really
deep tree um and then there's marov
chain uh Monte Carlo sampling where you
you're doing it very intelligently and
there's a lot I'm just starting to read
on these things but there's a lot of
depth to that and it it serves as a
basis
to uh sort of the alternative approach
to probabilistic programming from ARX
and fur um yeah it's quite quite
different yeah it's quite different but
but it
can I guess just one use case I I've
seen it on is where you condition on

each past time step and and like with RX
Ander you can um kind of go you can go
back you can trace every single step
through it so just like a high level of
this algorithm you sample you know one
step and you're like okay this is pretty
good and then based on you condition off
that you sample another and if you run
into a problem you go back to like the
last good State and then you try a
different
Direction um that's the high level
there's a lot of other really smart
optimizations on it um but it it can
allow for example like very effective
path finding um with minimal tree
exploration um yeah yeah
so oh yeah go ahead I was just gonna say
um that yeah that that's that sounds
really cool I I actually did my honors
project on this problem last year and I
have like a heuristic research thing but
the pain associated with that is ah you
still have to enumerate some amount of
St you know actions into the future for
some amount of policies and that gets
very painful very quickly um so you know
things like heuristic Tre search things
like I suppose some kind of Monte Carlo
something um they're sort of the only
things that I'm aware of uh for for
doing this kind of sampling based thing
if we're not going to do RX and fur type
stuff maybe we're getting a little bit
far field now but um this this is
let's just let's connect it back to to
the the basic problem and there's a
variety of methods to to solve it so the

fundamental issue is that if we have a
given number of affordances per time
step let's just say there's two like
left or right it it'ss a branching tree
then for a function of growing time
Horizon that you're considering there's
an exponentially growing number of total
policies because it's it's dou
so it's just like that's the problem of
Chess it's a problem of of getting
specific about
planning any other kind of control
method so then there's some some uh
structural methods that can amarate this
so one thing is like you might do a
parameter sweeper search like what time
Horizon do I really need to plan for
like do you need to plan 50 moves in
chess or what fraction of the the value
can you get get with five moves of
planning or something there's especially
an active inference there's nested
modeling so let's just say we were doing
a move every minute um we could and we
wanted to model
120 moves in the future well that might
be like a huge number of exponentially
exploding minutes or maybe it's only two
hours okay so then we have computational
methods for approximating these comp
um computations because we're going to
think of it as this kind of inference
problem and just because you can write
out the inference problem it's like okay
but but now the observations um it's a
4K video 60 frames per second and the
hidden state is going to be like the you
know so just because you can write the

equation simply doesn't mean like that
there's enough computational resources
on Earth to calculate it but the
equation still might be very simple so
then when that is happening there are as
kind of mentioned there's several
approaches ranging from different
sampling based approaches which attempt
to empirically approximate complex or
large or high-dimensional
distributions by speckling all over them
or taking smart paths through them and
then looking to reconstruct the entire
posterior so in this case we have like
an action prior is what we're taking
into this whole inference question we
want to update that action
prior and we want to update the
probabilities of taking different
actions according to their expected free
energy which is to say we want to like
UPR rank actions policies um that
contribute pragmatic value observing
future observations with our preferences
and and those that contribute epistemic
value okay so that's this question a few
few different ways to deal with it you
can structurally deal with it with
changing the structure generative model
you can do sampling based or variational
inference based approaches um let let
let's also get specific here's in
pmdp um tutorial
one this is what it looks like to to get
the expected free energy so this is like
operationalizing an equation like this
and this is um
programmed in the

PDP um imperative style so it's going to
give a sort of procedural code for how
you
calculate the expected free energy for
each
policy what expected free energy does is
it just takes in the policy vector and
for each policy considered separately it
calculates this value
so it's shown here and it's there's
other ways to compute it there's other
ways to write out the procedure to
compute this but this is that sort
of uh imperative style of pmdp where
calculations are described that take in
certain inputs combine them in certain
ways and result in certain outputs in
this case like the expected free energy
of moving left or
right um then there's the just more
again general questions for machine
learning about like uh policy rollouts
branching time active inference this was
described in several papers this is a a
method of heris tree search like rolling
out certain areas of the
tree
um that's oh leave
there but yes expected free energy
calculations
can be um having growing computational
complexity because if you want to be
dealing with something that's a large
policy space over long times then that's
a a pretty large
challenge yeah it's just it's
frustrating to have such a nice
beautiful equation that that quote

unquote sols it and you know it's it's
it's theoretically beautiful but it's um
it's very very difficult in practice for
I suppose non-toy problems to actually
do um analytic anyway that that was that
yeah that was my um main concern yeah I
mean um look at the computational
resources it and and uh compare it with
other
methods because that's true yeah it's
it's it isn't that each of these
expected free energy
calculations are necessarily that
resource intensive also they can be
paralyzed in the dispatch because
Computing the G for policy one and
policy 2 those are separate so that's a
highly paralyzable
step and and again it just depends on um
the overall like if you had a simulation
with a thousand nestmates making a
single step of action then the
computational bottleneck in that
simulation might not be G if you were
doing a single agent doing a thousand
time steps of planning maybe it would be
yeah that's probably how we actually get
around it is by by having many agents
kind of work together with with
relatively low time scales yeah
interesting to to think about pretty
pretty cool yeah Nest I mean and
multi-agents Andor nested models reduce
the need for explicit
planning because they have fewer
explicitly planned state or or like in
live stream 42 with the robot slam the
simultaneous localization mapping the

lower level model was the postural model
so extensive planning wasn't needed
there because it was like body posture
then the higher level model was
essentially the location of the
warehouse so then again planning was
rarely more than one or two nodes
deep so yeah that's because you had that
extra extra structure of of this
multi-layer uh model I suppose yeah yeah
yeah whereas trying to plan posture to
get across the warehouse would have
seemed
intractable oh awesome thank you Harris
yes exactly yeah then also this is sort
of this would be the the deeper the
deeper Tech angle would be like um as
discussed in some of the message passing
in Magnus cool dolls work in um live
stream
55 generalized free energy
encompasses variational and expected
free energy and it
enables another kind of nonplanning
based way to talk about reducing
uncertainty about Pol policy in the
future okay stockfish uses Alpha Beta
pruning search algorithm Alpha Beta
pruning improves Mini Max search by
avoiding variations that will never be
reached an optimal play because either
player will re rect the game yeah that's
That's a classic chess
algorithm and you can Implement in your
in your
script um any any you could say only
evaluate G for the even
policies or um any policy with three of

the same of um action selected in row
just remove it so there's there it's not
like we're procrustes and we're being
forced to do a
calculation these are just ways that you
can choose what you want to
calculate but it it also may be a lot
clear in a specific
case and each of these are just Matrix
operations so again if you're dealing
with massive matrices that you can't
core grain or or um discretize so like
you want you have a large Matrix and you
want to have continuous um
approximations and Etc it's like well
then you're setting up for a larger
problem but if your matrices are
discreet or
sparse then these are super fast
calculations and that's kind of the
amazing thing is there isn't any need
this is again the contrast with with
reward learning there's no secondary
like well let's calculate the reward
associated with the policy question mark
question mark question mark and then use
that to update policy it's like know
just directly calculate these exact
conditional probabilities that are
written here reweight your policy prior
into the policy posterior and then
select from
it or like shown in uh policy 5 here's a
more of a biological inspiration on that
um challenge with the dopamine
balance so here it's it's saying well
this is sort of habit on the left
here policy is being determined just by

The Habit so e is your is the policy
prior and so this is the kind of like
Thinking Fast system one and then here's
thinking slow system two where there's a
deliberative reweighting of
policy so then it then there's this
second level question which is when
should I
deliberate and if maybe only
deliberation has to be engaged in a tiny
fraction of cases and habits can be
carried through or updated and then just
reused okay chapter 7
so starts off again raise your hand or
or write if you have a question but it
starts off familiarizing with the
partially
observable Markov model or just the
hidden marov model just introducing okay
we're we're we're putting some space
with the a matrix between the
temperature in the room and the
thermometer reading so if it if it were
fully observable like here's the
location of the chest piece on the board
and this is the observation of the chest
piece on the board then it's a fully
observable marov
process there's an example with music
playing then action is brought
in action is not intervening in The
Hidden directly it's not it's not
reaching into the hidden state of the
room's temperature and modifying it it's
more like selecting an index card from
the B tensor and then selecting which
index card of B is going to get applied
to update the state so then it's like

there's one Matrix transition Matrix
that describes air conditioner on and
there's one that describes heater
on and then the air conditioner Maps
like 25 to 24 Maps 24 to 23 so it just
drops all the temperatures down by one
and the heater one maps all the
temperatures up by one so what action
does is Select which slice of B is going
to be used to multiply with the prior
state of s to reach the current state of
s so then then the question arises well
how do you choose which policy to take
and and here you could just use habit
like use just a fixed uh decision rule
or probabilistic Draw from a fixed uh
policy prior or there could be a another
way to select and that's
G one important figure for
G is
2.6 so here's G
again pragmatic value is now here on the
first term that it it it it doesn't
matter whether it's first or second term
pragmatic value epistemic value so an
important relationship to remember is if
you only have pragmatic
value then you have utility
maximization so in a situation where
there either is no epistemic value to
gain or it's just not considered the
special case of expected free energy
without epistemic value is basing
decision Theory expected utility Theory
conversely if you have no pragmatic
value again it's a situation that
doesn't have it or you're choosing not
to model it or you have like a uniform

preference over outcomes measurements
then you just get 2 three four five and
you get
Infomax so then having both it then
beets this question well how do you
balance both and that's where we see
some of the slightly
[Music]
um potentially inflated um language
around like dissolving the explore
exploit
dilemma does it does it dissolve it or
does it articulate it so that we have a
degree of freedom to balance it and and
adaptively explore Solutions but is that
dissolving
it so just just writing it out this way
like doesn't solve action selection in
general or for a specific
case but it's um a setting that allows
modeling a a tremendous range of
cases which
isn't an answer itself but that's kind
of the reason why there's so much
parameterization to do and why
fine-tuning the balance between these
two is an interesting
topic okay back to
seven
o
interesting it just like reloaded
differently didn't it
okay then action comes into play so
saying we took that music listening
example but this was a passive music
listening
example so now we're considering action
inference unified inference problem

sense making downstairs action selection
upstairs little bit more detail
revisiting equation 2.6
saying here's the functional that's used
to update the policy
distribution it's an energy based
functional that has a pragmatic and an
epistemic
component sometimes it gets little
tricky with like are we talking about
the negative free energy or the the the
negative of the free
energy suffice to say just check the
script and see how the values are being
stored and then just confirm it and
verify it like with what you know is
like a good policy in a limited
situation what you know is going to be a
inferior one relatively and just
calibrate it and reflect it and then add
it in the
documentation um so then we get to the
Tas a classic decision
example um here
we only have like one a
matrix but it turns out that you can
actually have multiple
A's that can have the same or different
shapes so it's kind of like parallel
A's
A1 is mapping the sense making
relationship between location and the
observation of the
location the reason why there's four
columns is because there's four
locations 1 2 3 four but the reason why
there's five rows is because there's
like five sense making outcomes because

the bottom location can either give this
l or the R so when we're in context one
which is to say that the bottom Q is
going to reveal an R then
deterministically being there will
reveal the r again this just begets the
question this is just all modeling well
how does the mouse come to know that
locations reflect their location or how
does it come to know the semantics of R
how does it come to associate r with the
right side like that that is the
question that's what then but then you
focus the map it's like saying well um
it's a subway map of the city but then
where where are the power lines it's
like there's a lot of ways to make
models of the situation the second a uh
component is mapping from location to
either the absence of either food or a
verse of stimuli or mapping the
probability of of measuring the either
um food or aversive
stimuli here's context one where the Q
tells you it's on the right and if you
go to the right 98% of the time there's
the food here here's context two so you
see a swap now the one is
here if you go to the Q you get the
L and now the 98 is there but the 98 was
here and the one was
there so that's the kind of like Matrix
surgery that in the nitty-gritty a lot
of this gets down to it's like what are
the rows and The Columns of
a now we into
B here these are um as PDP goes into in
a lot of

depth you have transition for um
controllable Transitions and
uncontrollable
transitions uncontrollable transitions
are one where the agent's policy
selection has no efficacy so it's not
really a control setting it's more just
like a Time series unfolding it's kind
of like a passive inference
element and then the B Matrix describes
the agent's beliefs about its efficacy
of action again this is another question
well what if the um driver believes that
pushing on the gas will will have this B
Matrix effect on their acceleration but
actually the car has a different
acceleration if that were the question
you were modeling then you might look at
a situation where there's a misat match
between the believed efficacy and the
actual efficacy but the simpler setting
is one where like what the agent
believes an action will do is what it
actually does but that's a major degree
of freedom in modeling is how do agents
have accurate representations of sense
making and
action then so it's just it's the ABCs
it's like Sesame Street
then it goes into describing
preferences just like reflecting the one
and the two of the
a A1 is location A2 is basically
food so C1 corresponds to location
there's a negative one kicking it off
the starting blocks and then it has a
flat locational preference this gets
into

interpretability and AI alignment and
safety in all these topics because here
we can say it's not learning a proxy
measure there is no pragmatic value
being contributed by moving up in the
teamas OR moving to the right we can
interpret this model and say it it does
have an e to the 6th pragmatic
preference for getting food that is
contributing to pragmatic policy these
ways you could even do Trace back and
you could look at how the expected free
energy values were contributed to by
specific pragmatic and epistemic value
components and in the simple case it's
all just Matrix math so that's the
preferences here again the thing to kind
of look out for is are the preferences
being uh encoded
in exponent form already in which case a
six is e to the 6 fold preference or you
might choose to in your program or in
the program that you're using six might
reflect a six-fold preference and then
and then under the hood it gets
exponentiated and just dealt with a
way a b c d here D1 location it begins
with perfect knowledge that it's 100% in
the starting location D2 that's about
the food so D2 is saying so it's about
the food
context it has an equal probability
here this is also alluding a little bit
to the learning by
counting so this one
one is like it's one half of one one so
it's 0.5 0.5 but learning by counting
what that would look like would be every

time that you um so this is trial by
trial learning so let's just say that it
was on the right side then you could
change this to one3
1 comma 2 so then that that would be
equivalent to saying 3 and
67 and then now let's just say it went
to write again it' be 1/4 1 comma 3 then
it' be 0.
25.75 so that's the learning by counting
because you basically just increment
what you've
seen and then if it's 1 million 1
million then it's like a higher
Precision belief because seeing one more
is only going to move your update a tiny
bit the Taz is again mobilized to focus
a little bit on exploration
exploitation setting and information
foraging continuing on the epistemic
value theme they move into an is
Paradigm but it's separated by from
where it's brought up by several Pages
there's a box that talks about Precision
just in a general
sense here's a seccade model there are
icade models with discrete and
continuous
elements then a discussion on
learning if there's no hyper prior
there's no hyper parameter for a given
parameter of Interest the base case is
that it's fixed and not
learnable so when we're looking at the
teamas we're looking at like this a
matrix this a matrix is being used to
run forward inference like to run sense
making and um to the extent that it has

values that contribute to it it also
contributes epistemic values to
policy but the a itself is not learnable
it is not an object of inference it's
not updated the values are not updated
in the base case but this describes how
you could make a basian model where
there is updating of the a the B
Etc this is the learning by Counting
more unpacking of equation
2.6 more different decompositions of
expected free
energy and demonstrating that in this
maze
paper it's bringing in a lot of
topics structure learning viewing
alternative structural models including
with different numbers of parameters as
essentially a policy choice of the
modeler so then that allows you to have
the metab basian approach to basian
selection and then here hierarchical
nested
modeling and a nested model
so seven is kind of a
lot but it covers starting from a a
passive listening
example all the way on through action
and then some of these uh doing a little
bit of showing not telling on unifying
cognitive modeling and treat in
different cognitive phenomena in terms
of free
energy does anyone have a question or we
can look at some of the written
questions or please please add many
questions to these
staples nothing for me right now sorry

this was some I think this was Eric
sounds from several years
ago it's like even taking another level
of meta there is no
dilemma explore exploits already sted in
the clearest
way yeah I mean there's there's
trade-offs of that kind everywhere
efficiency and robustness and so on so
it's not like you have to choose one of
the other or that if you choose one you
can't choose the other yeah I suppose
you know if he gives you a principled
way perhaps of deciding which one you
should trade off at any given time
maybe that's what it's meant by it
solves it but yeah it's it's a question
for interpretation of what it means to
solve it's like saying because we have a
volume knob you don't have to worry
about the
volume it's like no that means the sound
engineer is worried about the
volume it's like he has a principled way
of controlling it yeah and and there's
an interpretable simple way so for
example you could have a uh an agent
with a fixed epistemic value so kind of
certain uncertainties known unknowns
about the world and then there's a
policy on pragmatic value policy on the
preferences like I scale up and down how
much I am excited about food but
studying is always the same amount so
then when I want to focus on studying I
just kind of Turn Down how much I care
about food and or turn it up if I want
food or you could say I don't have

control over C in this
case and then you could explore well but
could we tune the epistemic side to
modify essentially the Curiosity from
lower to higher levels to affect the
same
concept but whether you're modifying the
the scale of the pragmatic value or the
scale of the epistemic
value like you still need to then have
this question well what what would make
an Adaptive approach to navigating that
tradeoff let's just see what else we
have what I cannot create I do not
understand some people feel again more
funny comments that people have
written the the mild
answer okay the mild answer you need to
implement the algorithms and see how
they perform well the cool thing is
there's
notebooks like in RX Ander and pmdp
where even if you don't have any program
um language installed locally like just
in the browser you can hit play and get
the active inference agent that may help
certain people if you'd like to
contribute to those efforts and improve
the notebooks that's a great way because
a lot of people will benefit from it
stronger answer
here okay let's see math and theory and
hand waving how the brain might do thus
and so are all weak
te the only way to really prove the idea
is to build artifacts that do
interesting things especially things
that other methods cannot accomplish

that's sort of the embodied
robotics JF cler moral comp moral
computation
method rows and columns these are great
understanding checks
like if if you look at the rows and The
Columns of all the
matrices
um that are
reflected in figure
7.3 you basically understand the
discrete time active inference
model because that's what it is each of
these are matrices or
tensors and then the operations are
essentially manipulations or
multiplications on these
matrices so that's a very deflationary
approach to active inference but it's
super informative because it's like
there's not like some other magic layer
that then is is applied
here similarities and differences
between prediction error and free energy
gradients that's a good one does that
does that show up in uh chapter seven
like um prediction error
specifically it looks like it's in the
caption some people use them including
sometimes even possibly authors who who
know better slash differently but and
and there's so it's you know words words
words etc etc etc but a prediction error
is denominated in the units of what the
prediction is about like if I predict
that it's going to be 30 degrees and
then it's 32 degrees the prediction
error was two

degrees
surprise is denominated in units of
information Theory like the knots or the
bits depending on base e or base 2 and
then for free energy is is a unitless
quality a quantity that that's that's
just a
calculation so you could have a free
energy relative free just like you could
have a maximum likelihood or a
likelihood value for 30 degrees and 32
degrees you could have a free energy
value for 30 and 32 degrees as part of
the model fitting with respect to a
generative model but once you get into
talking about free energy or surprise
you're talking about with reference to a
given generative model whereas pred
ition error in the kind of narrowest
sense also it it does entail the model
making the prediction but it is
denominated in the units of what the
prediction is
about
h i I like that the talking about what
the prediction is about because I think
the two aren't inherently so related for
example like if I'm trying to improve my
pragmatics or something like that in the
sense that maybe I'm playing a fighting
game and the uh if I'm understanding
this correctly the actual loss what the
model is reflecting back is that I'm
practicing my combos I'm practicing some
inside of that that is going to be some
instrumental thing that I want in order
to improve at the game but in me doing
that I actually start to lose more often

so the actual expected loss the actual
thing that I expect to see is me
improving my model of how the game works
while the actual reality of it is that I
losing at the game more I think that's a
interesting
thing oh okay so it's like a game where
um you're or it's like I wonder what
will happen if I play this
differently and then you might be
learning
more even though the so kind of leaning
into the epistemic
side like yeah what if I what if I try
to play with what if with just this
weird chess
strategy with a longer term pragmatic
value or
something that's an interesting point
about the um prediction ER is being
denominated in specific units because
it's it's called you know it's about
some qualitative phenomena out there
whereas the VF is or you know for it's
just more about the fit of your
beliefs that's that something I hadn't
considered
yeah free energy value is a lot like a
model likelihood value for several
reasons first off model likelihood
saying maximum likelihood
ml is minimum
surprise so when we say that free energy
is kind of bounding
surprise like in the special case or or
in the complete case the free energy is
one and the same as the surprise
minimization which would make it the the

the evidence maximization or in the sort
of more General approximation case it's
it's heading closer and closer towards
minimizing surprise so it's heading
closer and closer to maximizing model
evidence but key similarity with
likelihood
estimates uh or free energy estimates or
like sum of squares is they're not
comparable across different
settings like the sum of squares the L2
Norm that's used in linear
regression that depends on the number of
data points you add another data point
and even if it doesn't change the
regression line it's going to like it it
it changes the sum of the squares of the
residuals but that so that's why the sum
of squares are not compared for like
data sets with different numbers of
value so free
energy will also C
EG sum of
squares
likelihoods they're they're like a
statistical
diagnostic I mean these are summary
variables on
maps again this is part of the the
deflationary educational Elements which
is like once free energy is seen as the
calculation that it's described as in
the text
ttbook it puts some of the discussions
around real systems and their free
energy into a different
light because it'd be like well would
would this person also argue that the

sum of squares of the regression between
height and weight is a real thing in the
world okay would they argue that the
free energy value of the relationship
between height and weight is a real
thing in the world so then what is the
claim
any last uh thoughts or or or what
people plan to
like add for questions or work on for
the coming few weeks
I guess the only thing is um the this
this the thing that I find is natural to
follow on from this discussion is the
kind of stuff I brought up earlier about
um particular ways to deal with the
explosion in the policy
space that's that's kind of where I feel
like uh this this whole chapter like
where you would want to go after this
let's say so that's that's one thing I
I'm very interested in is uh trying to
trying to further those huris means of
you know navigating those very large
spaces yep that's important topic um I
mean also when uh build things out and
whether timing them like just using
logging and timing functions in the
script or or using other ways like just
explore and be like yeah what what is
what does the runtime look when I plot
the runtime for time for 1 2 3 4 five
step time
Horizon like what does that look like
and then this model stream
6.1 this is really interesting work with
the branching time active inference and
this is some of the only still actually

some of the only computational
complexity like Big O
notation I don't know why it's not
loading but for active so so look at
that if you're interested in in
computational complexity
estimates but that's another great
advantage that with our X infer and
everything we will um realize which is
like we let's just say that we knew okay
temperature it's a number between one
and
100 observations number between one and
100 this has this defined dimensionality
this has this defined dimensionality in
the discreet time we could say exactly
the RAM and the CPU and all these other
attributes about the model
statically so then you could anticipate
resourcing
deterministically even for a
probabilistic
model there's a there's a lot of things
like that and then you can say well now
we have three ways to implement this we
could we could do the full policy roll
out you know that's going to be
1999 or here's the you know half off but
we only we only sample half the
policies uh Daniel would you be able to
link that that Big O paper talk you'd
mentioned it's actually something
curious
ranching model stream there's actually
two papers I believe there's the there
is the original paper and then there's a
followup which is the kind of Big O
stuff I

think
yes L link that or just add it you know
any and then anywhere we search we'll
find it but yeah that that that's cool
and uh yeah that was like sort of our I
think just last week in ARX and F we
were discussing like how much other
computation is there like what is the
overhead what and then what like if the
payload is this sort of like
hypothetical minimum computational
complexity of just storing the matrices
and doing the operations and then it's
like how much is the computational
burden of the message
passing and like what are the situations
where the Matrix math has a given
computational complexity and then the
message passing has like this or that
right yeah that's a big open question to
me um hope read more about where it's
covered just because there's so many
different message passing scenarios
yes okay so
um all right paper
two that's the the the sort of theory
and then paper one is the uh complexity
time complexity class
analysis I
believe cool thank you f cool well um
see you all either uh at this time like
next week or later today Andrew will
be
uh facilitating I
believe so enjoy um talk to you soon and
not anything till then all right bye bye
by